Protocol I8B-MC-ITRU A Study of LY900014 in Participants With Type 2 Diabetes Mellitus
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Protocol I8B-MC-ITRU
A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog Following a Single Dose in Patients with T2DM

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LY900014

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Table of Contents

A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog Following a Single Dose in Patients with T2DM

Section **Page**

Protocol I8B-MC-ITRU A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog Following a Single Dose in Patients with T2DM 1
Table of Contents 2
1. Protocol Synopsis 8
2. Schedule of Activities 11
3. Introduction 15
3.1. Study Rationale 15
3.2. Background 15
3.3. Benefit/Risk Assessment 16
4. Objectives and Endpoints 18
5. Study Design 19
5.1. Overall Design 19
5.1.1. Inpatient Dosing Visits 21
5.2. Number of Participants 22
5.3. End of Study Definition 22
5.4. Scientific Rationale for Study Design 22
5.5. Justification for Dose 23
6. Study Population 24
6.1. Inclusion Criteria 24
6.2. Exclusion Criteria 26
6.3. Lifestyle and/or Dietary Requirements 28
6.3.1. Meals and Dietary Restrictions 28
6.3.2. Caffeine, Alcohol and Tobacco 28
6.3.3. Activity 29
6.3.4. Washout Period 29
6.4. Screen Failures 29
7. Treatment 30
7.1. Treatment Administered 30
7.1.1. Packaging and Labelling 30
7.2. Method of Treatment Assignment 30

7.2.1. Selection and Timing of Doses 31
7.3. Blinding 31
7.4. Dose Modification 31
7.5. Preparation/Handling/Storage/Accountability 31
7.6. Treatment Compliance 32
7.7. Concomitant Therapy 32
7.8. Treatment after the End of the Study 32
8. Discontinuation Criteria 33
8.1. Discontinuation from Study Treatment 33
8.1.1. Discontinuation of Inadvertently Enrolled Patients 33
8.2. Discontinuation from the Study 33
8.3. Patients Lost to Follow-up 34
9. Study Assessments and Procedures 35
9.1. Efficacy Assessments 35
9.2. Adverse Events 35
9.2.1. Serious Adverse Events 36
9.2.1.1. Suspected Unexpected Serious Adverse Reactions 36
9.2.2. Complaint Handling 37
9.3. Treatment of Overdose 37
9.4. Safety 37
9.4.1. Laboratory Tests 37
9.4.2. Vital Signs 37
9.4.3. Electrocardiograms 38
9.4.4. Physical Examinations 38
9.4.4.1. Body Weight 38
9.4.4.2. Hip and Waist Circumference 38
9.4.5. Safety Monitoring 38
9.4.5.1. Hepatic Safety 39
9.4.5.2. Glucose Monitoring 39
9.4.5.3. Severe Hypoglycaemia 40
9.4.6. Injection-Site Assessments (Local Tolerability) 41
9.4.6.1. Pain Measurements using the Visual Analogue Scale 41
9.5. Pharmacokinetics 41
9.5.1. Bioanalysis 41
9.6. Glucodynamics (Euglycaemic Glucose Clamp) 42
9.7. Immunogenicity Assessments 43
9.8. Genetics 43
9.9. Exploratory Biomarkers 44

9.9.1. C-Peptide 44
9.10. Health Economics 44
10. Statistical Considerations and Data Analysis 45
10.1. Sample Size Determination 45
10.2. Populations for Analyses 45
10.2.1. Study Participant Disposition 45
10.2.2. Study Participant Characteristics 45
10.3. Statistical Analyses 45
10.3.1. Safety Analyses 46
10.3.1.1. Clinical Evaluation of Safety 46
10.3.1.2. Statistical Evaluation of Safety 46
10.3.1.3. Statistical Evaluation of the Intensity of Injection-Site Pain 46
10.3.2. Pharmacokinetic Analyses 46
10.3.2.1. Pharmacokinetic Parameter Estimation 46
10.3.2.2. Pharmacokinetic Statistical Inference 47
10.3.3. Glucodynamic Analyses 48
10.3.3.1. Glucodynamic Parameter Estimation 48
10.3.3.2. Glucodynamic Statistical Inference 48
10.3.4. Evaluation of Immunogenicity 48
10.3.5. C-Peptide 49
10.3.6. Interim Analyses 49
11. References 50

List of Tables

Table		Page
Table ITRU.1.	Objectives and Endpoints	18
Table ITRU.2.	Treatments Administered	30

List of Figures

Figure **Page**

Figure ITRU.1. Illustration of study design for Protocol I8B-MC-ITRU. 21

List of Appendices

Appendix		Page
Appendix 1.	Abbreviations and Definitions	51
Appendix 2.	Clinical Laboratory Tests	56
Appendix 3.	Study Governance, Regulatory and Ethical Considerations	57
Appendix 4.	Hepatic Monitoring Tests for Treatment-Emergent Abnormality	60
Appendix 5.	Blood Sampling Summary	61

1. Protocol Synopsis

Title of Study:
A Study to Evaluate the Pharmacokinetics and Glucodynamics of LY900014 compared to Humalog Following a Single Dose in Patients with T2DM

Rationale:
LY900014 is an ultra-rapid-acting insulin lispro formulation with increased early absorption compared to commercially available insulin lispro formulation (Humalog®; Eli Lilly). LY900014 aims to more closely mimic the physiological prandial insulin secretion pattern, which may, more effectively, control postprandial glucose excursions.

The aim of this study is to compare the insulin lispro pharmacokinetic (PK), glucodynamic (GD), safety and tolerability profiles of insulin lispro following administration of either LY900014 or Humalog during a euglycaemic glucose clamp in patients with type 2 diabetes mellitus (T2DM).

Objectives/Endpoints:

Objectives	Endpoints
Primary	
To evaluate the PK of insulin lispro following administration of a single 15 U subcutaneous (SC) dose of LY900014 compared to Humalog in patients with T2DM	Time to early half-maximal drug concentration (early 50% t_{max}) and area under the concentration versus time curve (AUC) from time zero to 30 minutes (AUC[0-30min])
Secondary	
1. To compare the euglycaemic clamp GD of insulin lispro following administration of a single 15 U SC dose of LY900014 or Humalog in patients with T2DM	1. Time to early half-maximal glucose infusion rate (early 50% tR_{max}), total amount of glucose infused (Gtot) over 30 minutes ($Gtot_{[0-30min]}$) and time to onset of insulin action (t_{onset})
2. To evaluate the tolerability of LY900014	2. Adverse events (AEs) and injection site reactions

Summary of Study Design:

Study I8B-MC-ITRU is a randomised, patient- and investigator-blind, 2-treatment, 2-period, 10-hour euglycaemic clamp crossover study in patients with T2DM.

Treatment Arms and Planned Duration:

Patients will be randomised to 1 of 2 treatment sequences according to the actual randomisation table provided to the site. All treatment procedures will be inpatient stays of 2 days per period and require a maximum 5 weeks to complete 2 periods. A minimum period of 72 hours is required between each 10-hour clamp visit during which patients will resume their normal insulin treatment following instructions from the site investigator. The follow-up or early discontinuation visit should occur at least 14 days after the last dose of the study drug.

LY900014: Single 15 U dose

Humalog: Single 15 U dose

Number of Patients:
Up to 42 patients may be enrolled to ensure at least 34 patients complete the study.

Statistical Analysis: Primary statistical analyses will be conducted on the set of patients who complete at least the first period of treatment. Supportive analyses will be done on the key parameters for the patients who complete all treatment periods. Statistical significance will be achieved when the p-value for a test is less than 0.05.

Safety: Safety analyses will be conducted for all enrolled patients, whether or not they completed all protocol requirements. Safety assessments including, but not limited to vital signs, safety laboratory parameters, investigational product, procedural and protocol AEs will be captured and summarised using descriptive statistical methodologies.

Pharmacokinetics: PK analyses will be conducted on all patients receiving at least 1 dose of study drug and have evaluable PK. PK analyses will be conducted using standard noncompartmental method of analysis. PK parameters will be assessed using free serum insulin lispro.

Log-transformed PK parameters for insulin lispro will be evaluated to estimate least-squares geometric means, ratios of geometric means between LY900014 and Humalog and their corresponding 95% confidence intervals (CIs) using the mixed-effects model that includes treatment, sequence and period as fixed effects and patient within sequence as a random effect.

The same model without log transformation will be used for the analysis of the PK time parameters (early 50% t_{max}, time to late half-maximal drug concentration [late 50% t_{max}], time to maximum observed drug concentration [t_{max}], and half-life associated with the terminal rate constant in non-compartmental analysis [$t_{1/2}$]). Least-squares means, treatment differences in least-squares means and the corresponding 95% CIs for the treatment differences will be estimated from the model. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

The analyses described above will also be performed on the population of patients who completed and had evaluable PK data for both study periods.

Glucodynamics: Glucodynamic assessments (GD) will be determined from the glucose clamp procedure, where the glucose infusion rate (GIR) over time will be used as a measure of insulin effect. Glucodynamic analyses will be conducted on those patients who complete at least 1 clamp procedure. A locally weighted scatterplot smoothing function will be applied to all individual GIR versus time profiles in each treatment group and/or period. The fitted data for each patient will be used to calculate the following GD parameters:

The GD statistical model will be the same as the model used for the analysis of the PK parameters. The following variables will be log transformed prior to analysis: maximum GIR (R_{max}), Gtot, Gtot over 30minute (Gtot[0-30min]), and Gtot over 1 hour (Gtot[0-1h]).

The same model without log transformation will be used for the analysis of the GD time parameters (t_{onset}, time to R_{max} (tR_{max}), time to half-maximal GIR before tR_{max} (early 50% tR_{max}), time to half-maximal GIR after tR_{max} (late 50% tR_{max}). Least-squares means, treatment differences in least-squares means and the corresponding 95% CIs for

the treatment differences will be estimated from the model. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

The analyses described above will also be performed on the population of patients who completed and had evaluable GD data for both study periods.

2. Schedule of Activities

Study Schedule Protocol I8B-MC-ITRU

	Screening	Euglycaemic Clamp Procedure Periods 1 and 2		Follow-up/Early Discontinuation	Instructions/Comments
Procedure	Up to -28 days	Day -1	Day 1	At least 14 days after discharge	
Informed consent	X				Informed consent must be signed at least 1 day prior to screening procedures
Admission to CRU		X			Admission to site at early evening
Overnight fast			X		Patients are expected to fast for approximately 8 hours prior to screening and approximately 8 hours before each dose until the end of the glucose clamp procedure
Hip and waist circumference		X			Period 1 only. Measurements taken in triplicate
Height	X				
Weight	X	X			Period 1 only
Medical history and physical examination	X				
Medical assessment		X	Predose and before discharge from CRU for each period	X	Medical assessment includes medical review and targeted examination as appropriate
Vital signs (supine): body temperature, blood pressure and pulse rate	X		Predose and approximately 10 hours postdose (at the end of the clamp procedure)	X	
12-lead ECG	X		Predose and approximately 10 hours postdose (at the end of the clamp procedure)	X	Single ECGs will be collected for safety
Clinical laboratory tests	X		Predose for Period 1 only	X	Fasting laboratory test for screening, Period 1 only as well as for follow-up or early discontinuation. To

					monitor patient safety, additional tests may be performed at the discretion of the investigator as needed throughout the study. Screening laboratory tests are analysed at a local facility
Pregnancy test	X	X		X	For female patients of childbearing potential only. A serum pregnancy test will be performed at screening, analysed at a local laboratory, and urine pregnancy tests will be performed at Day -1 of Period 1 and at follow-up and processed on site
Run-in and stabilisation of glucose			X		To start approximately 1 to 6 hours prior to study drug dosing to achieve blood glucose target of 100 mg/dL (±20%) through the use of infusion of glucose (dextrose solution) or insulin glulisine. See Section 5.1.1 for rescheduling.
Study drug administration			X		Time 0: Study drug administration
Injection-site local tolerability assessments			0, 60, 240 and 600 minutes postdose		Time 0: Assessments of injection-site local tolerability will occur immediately following the injection
VAS Pain assessment			0, 20 and 60 minutes postdose		Time 0: Assessments of injection-site pain will occur immediately following the injection
Insulin lispro PK sampling			0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 90, 120, 150, 180, 240, 300, 360, 420, 480, 540 and 600 minutes		Time 0: Sample to be taken immediately before study drug administration. Sampling times are relative to the time of study treatment administration (t = 0).
Blood glucose sampling (euglycaemic clamp)			Blood glucose monitoring every minute starting from the run-in and throughout the duration of the clamp until 10 hours		

			after dosing		
Blood glucose sampling (Super GL)			Approximately every 30 minutes throughout the euglycaemic clamp procedure		
C-peptide sampling			0, 30, 60, 120, 180, 240, 300, 360, 420, 480, 540 and 600 minutes		Time 0: Sample to be taken immediately before study drug administration. Sampling times are relative to the time of study treatment administration (time = 0).
Pharmacogenetics sample			Predose for Period 1 only		Refer to sample collection instructions provided by the sponsor
Immunogenicity sample			Predose for Periods 1 and 2	X	Additional samples may be collected if the investigator considers there is a possibility that an AE is immunologically indicated
Discharge from CRU			X		Patients may be discharged after completion of all study procedures on Day 1. Investigator's discretion to discharge on Day 2

Abbreviations: AE = adverse event; CRU = clinical research unit; ECG = electrocardiogram; PK = pharmacokinetics; VAS = visual analogue scale.

Note: The site should schedule activities as appropriate. In cases where several study procedures are scheduled at the same time, the order of priority will be as follows: PK samples including blood sampling for blood glucose and laboratory samples per protocol nominal times. ECGs and vital sign measurements should be scheduled before but as close as possible to the PK sampling times. VAS and injection-site assessments can be done after PK sampling.

3. Introduction

3.1. Study Rationale

LY900014 is an ultra-rapid-acting insulin lispro formulation that has shown an increased early absorption compared to commercially available insulin lispro (Humalog®; Eli Lilly). LY900014 aims to closely mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions.

The aim of this study is to compare insulin lispro pharmacokinetic (PK) profiles and glucodynamic (GD) profiles following administration of either LY900014 or Humalog during a euglycaemic glucose clamp in patients with type 2 diabetes mellitus (T2DM).

3.2. Background

The insulin analogue insulin lispro (Humalog®) has been shown to be absorbed more quickly than regular human insulin (Humalog package insert, 2015). In healthy volunteers given subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 U/kg, peak serum levels were seen 30 to 90 minutes after dosing (Humalog package insert, 2015). However, the general consensus is that rapid-acting insulin is still not rapid enough to match carbohydrate absorption profiles, which limits efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the PK/GD of insulin analogues so that they have an even faster onset to better match carbohydrate absorption.

LY900014 represents a new formulation that contains insulin lispro, treprostinil, citrate and other excipients. This formulation involves the novel use of a microdose of treprostinil (CCI [redacted]) as an excipient to enhance the absorption of insulin lispro by local vasodilatation rather than as an active pharmaceutical ingredient to elicit a systemic effect. Treprostinil is a prostacyclin analogue, administered either through inhalation (CCI [redacted]), as an intravenous (IV) infusion or as a continuous SC administration for the treatment of symptomatic pulmonary arterial hypertension (PAH) and has been approved in the US since 2002 and in Germany since 2007 (CCI [redacted]). Sodium citrate, an excipient that speeds insulin absorption (likely in part by enhancing vascular permeability), is also included in the formulation to further enhance the absorption of insulin lispro. Each of the other excipients (such as sodium citrate and magnesium chloride) in the LY900014 formulation is listed in the US Food and Drug Administration (FDA)'s Generally Recognized as Safe Food Additives database and in the FDA's Inactive Ingredients in Approved Drugs database. Furthermore, the excipient concentration in LY900014 is within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Safety and tolerability of LY900014 have been demonstrated in approximately 71 healthy subjects in 3 previous clinical studies, across a dose range of 7.5 to 30 U. All 3 studies were Phase 1, randomised, subject-blind studies in which the PK and GD of insulin lispro from LY900014 and Humalog were evaluated during a euglycaemic glucose clamp following SC administration. The total insulin lispro exposure and GD effect were similar for LY900014 and

Humalog; however, LY900014 demonstrated a faster and earlier insulin lispro absorption compared to Humalog.

In addition, data from two Phase 1b studies showed LY900014 were well tolerated in patients with type 1 diabetes mellitus (T1DM; 30 patients) and in patients with type 2 diabetes mellitus (T2DM; 30 patients) using multiple daily injections (MDIs). There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Small numbers of treatment-emergent adverse events were reported, and there were no notable increases in these events in relation to any of the previous LY900014 formulations compared to those in relation to Humalog.

More information can be found in the Investigator's Brochure (IB) of LY900014.

3.3. Benefit/Risk Assessment

Study I8B-MC-ITRU (ITRU) will not offer any direct benefits to the patients participating in the study. Data from studies with healthy subjects and patients with T1DM and patients with T2DM have shown that LY900014 is well tolerated with an adverse drug reaction profile consistent with that of Humalog.

Potential risks associated with LY900014 are derived from the known risks of insulin lispro (Humalog). These potential risks include hypoglycaemia, hypersensitivity reactions (localised allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy) and peripheral oedema (Humalog package insert, 2015).

Notably, across all doses in the studies that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, CCI [redacted]. The exposures of treprostinil in LY900014 in this trial are expected to be undetectable compared to those observed in the dose ranges previously explored with SC bolus administration of treprostinil and are expected to be substantially lower than those observed in the treatment of PAH.

In preclinical safety pharmacology and toxicity studies, or clinical pharmacology studies involving LY900014 or treprostinil alone, other than known risks associated with Humalog and Remodulin, no additional risks were identified. No known potential risks are associated with the use of small amounts of treprostinil (CCI [redacted]) in the LY900014 formulation.

Additionally, local and systemic toxicity profiles of Humalog and Remodulin do not suggest the potential for additive or synergistic toxicity.

In order for patients to participate in this study, they must discontinue use of certain oral antidiabetic (OAD) medications 2 weeks prior to Period 1, Study Day -1 (see Section 5.1). Additionally, patients are required to discontinue their basal insulin prior to Study Day -1 according to the guidance in Section 5.1. Appropriate measures will be taken to minimise the risk of hyperglycaemia (see Section 6.3.4).

Following administration of the study insulin (Humalog or LY900014), patients will receive IV glucose infusion at a variable rate to maintain euglycaemia up to 10 hours after insulin lispro administration. The aim of the clamp procedure is to maintain blood glucose within the normal glycaemic range. In addition, the clamp is performed while the participant is inpatient and under the investigator's supervision. These considerations should minimise the risk of hypoglycaemia in patients participating in Study ITRU.

More information about the known and expected benefits, risks, SAEs and reasonably anticipated AEs of LY900014 is to be found in the IB.

4. Objectives and Endpoints

Table ITRU.1 shows the objectives and endpoints of the study.

Table ITRU.1. Objectives and Endpoints

Objectives	Endpoints
Primary	
1. To evaluate the PK of insulin lispro following administration of a single 15 U subcutaneous dose of LY900014 compared to Humalog in patients with T2DM	1. Early 50% t_{max} and AUC(0-30min)
Secondary	
1. To compare the euglycaemic clamp GD of insulin lispro following administration of a single 15 U SC dose of LY900014 or Humalog in patients with T2DM	1. Early 50% tR_{max}, $Gtot_{(0-30min)}$ and t_{onset}
2. To evaluate the tolerability of LY900014	2. AEs and injection-site reactions
Exploratory	
1. Explore the formation of anti-drug antibodies to insulin lispro	1. Anti-insulin lispro antibodies
2. To assess C-peptide levels following administration of LY900014 and Humalog	2. C-peptide concentration

Abbreviations: AE = adverse event; AUC(0-30min) = area under the concentration versus time curve from time zero to 30 minutes; early 50% t_{max} = time to early half-maximal drug concentration; early 50% tR_{max} = time to early half-maximal glucose infusion rate; GD = glucodynamics; $Gtot_{(0-30min)}$ = total amount of glucose infused over 30 minutes; PK = pharmacokinetics; SC = subcutaneous; T2DM = type 2 diabetes mellitus; t_{onset} = time to onset of insulin action.

5. Study Design

5.1. Overall Design

ITRU is a Phase 1, randomised, patient- and investigator-blind, 2-treatment, single-dose, 2-period, 10 hour glycemic clamp, crossover study that evaluates the PK and GD of insulin lispro following administration of a single 15 U SC dose of either LY900014 or Humalog in patients with T2DM. Study ITRU may be conducted at 1 or more sites. Figure ITRU.1 illustrates the study design. Patients will be required to attend at the clinical research unit (CRU) on at least 5 occasions as noted in Study Schedule (see Section 2):

- informed consent
- screening
- inpatient study Periods 1 and 2
- follow-up or early discontinuation

Patients will be required to visit the CRU to sign the informed consent form (ICF). Once the ICF is signed, patients will return to the CRU in a fasted state to participate in the screening procedures. Screening procedures may be completed up to 28 days prior to randomisation.

Prior to CRU admission on Day -1 of Period 1, patients with T2DM receiving OAD medication should discontinue their medication according to the following guidance (patients can continue taking metformin throughout the study):

- At most, 2 weeks prior to Day -1 of Period 1, patients should discontinue dipeptidyl peptidase (DPP)-IV inhibitors, sulphonylureas and sodium-glucose co-transporter (SGLT)-2 inhibitors

Prior to each dosing visit, patients with T2DM on multiple daily insulin injections have to discontinue their basal insulin, according to the following guidance:

- For patients using insulin glargine U300, the last injection should occur no later than 72 hours prior to dosing.
- For patients using insulin detemir or glargine, the last injection of insulin detemir or glargine should occur no later than 48 hours prior to dosing.
- For patients using neutral protamine Hagedorn (NPH) insulin, insulin mixtures or other intermediate-acting insulin, the last injection of NPH insulin or other intermediate-acting insulin should occur no later than 24 hours prior to dosing.
- Any injection of more than 6 U of short-acting insulin should occur between 11 and 6 hours prior to dosing. Any injection should occur no later than 6 hours prior to dosing.

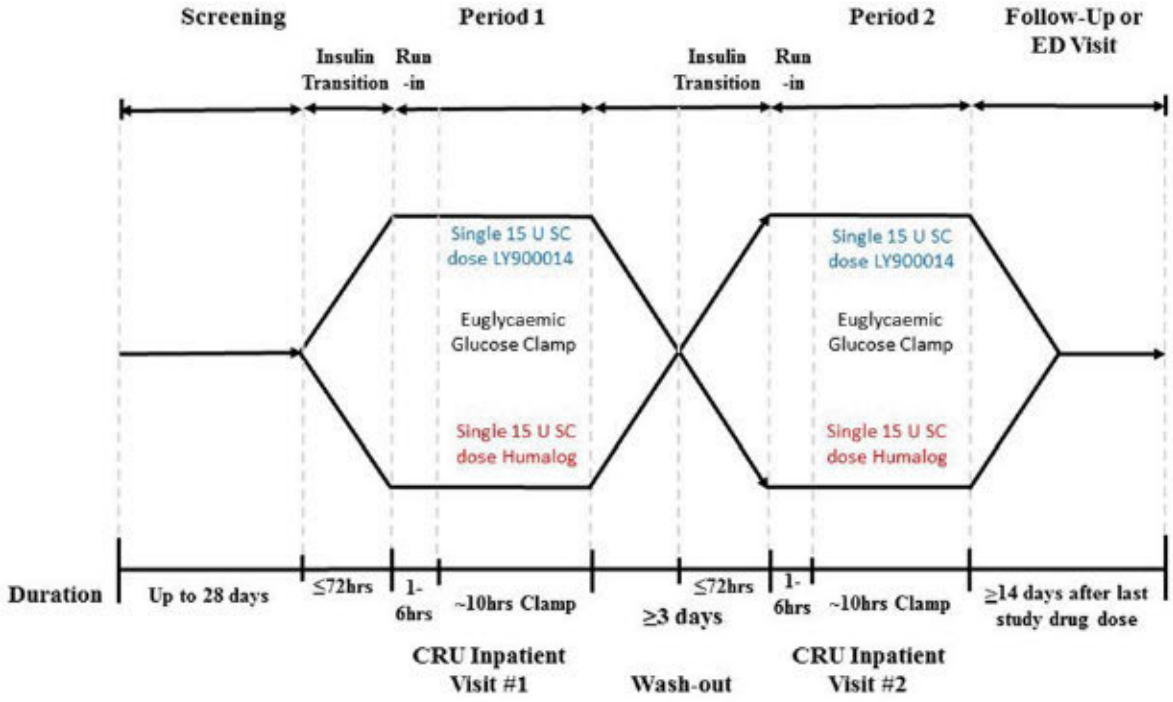
All patients will be instructed to perform at least 4-point daily self-monitoring plasma glucose (SMPG), as well as continue their short-acting insulin during insulin transition period (see Section 6.3.4). A diary will be provided to patients in order for SMPG results and prandial

insulin doses to be recorded during the insulin transition period. The patients' diary will serve as a source document.

After satisfying the study entry criteria, each patient will be randomised to 1 of 2 treatment sequences according to the actual randomisation table provided to the site. Each study period will be separated by a wash-out period of at least 72 hours. Each patient will participate in 2 inpatient visits; at each visit, the time-concentration and time-action profiles will be evaluated simultaneously during a euglycaemic glucose clamp of approximately 10 hours. The maximum duration allowed for both periods is 5 weeks. Briefly, the aim of the euglycaemic clamp is to maintain euglycaemia after the administration of a dose of insulin by means of variable glucose infusion. The variable glucose infusion maintains or "clamps" glucose to a constant euglycaemic target.

Patients will return to the CRU at least 14 days following last dose of study drug for follow-up or early discontinuation procedures.

Figure ITRU.1 illustrates the study design.



Abbreviations: CRU = clinical research unit; ED = early discontinuation; SC = subcutaneous.

Figure ITRU.1. Illustration of study design for Protocol I8B-MC-ITRU.

5.1.1. Inpatient Dosing Visits

The patient will check into the CRU on Day -1 of each period in the early evening. Patients on multiple daily insulin injections would have discontinued their basal insulin. All patients will receive an injection of short-acting insulin (e.g., insulin glulisine) before the start of a standardised dinner within the CRU. After dinner the patient is required to fast until the completion of the clamp procedure the following day. Consumption of food or beverages other than water later than approximately 22:00 on the evening before dosing is not allowed, with the exception of minor intake of rapidly absorbable carbohydrates (not more than 20 g), if necessary to prevent hypoglycaemia (if carbohydrates have been ingested, patients' blood glucose measurement will have to confirm that they are not hypoglycaemic). If hypoglycaemia (blood glucose level ≤3.5 mmol/L [≤63 mg/dL] or plasma glucose ≤3.9 mmol/L [≤70 mg/dL]) or more than 20 g of carbohydrates are needed to prevent hypoglycaemia less than 24 hours prior to dosing, the dosing visit can be rescheduled 1 to 7 days later. Each of the dosing visits can only be rescheduled once.

Prior to study treatment dosing, the patient will be connected to the clamp device for continuous glucose monitoring and the start of the baseline run-in period. Variable IV infusion of either glucose (20% dextrose solution) or insulin (insulin glulisine) will be started to reach a target blood glucose level of 100 mg/dL (±20%) during the run-in period. Once the target blood glucose level is attained and remains stable, with the predose activities as specified in the Schedule of Activities (Section 2) completed, the patient will receive a dose and undergo the clamp procedure (Section 9.6).

Upon completion of the clamp procedure, the patient may be offered a meal. It is planned that patients will be discharged from the CRU after completion of all study procedures on Day 1; however, at the investigator's discretion the discharge may be delayed to Day 2 for medical or logistical reasons. A minimum washout period of 3 days is required between each dosing period. During the washout period, patients will transition to their normal insulin treatment with guidance from the investigator. The maximum duration allowed for both study periods is 5 weeks.

More information on the clamp methodology can be found in Section 9.6.

Study governance considerations are described in detail in Appendix 3.

5.2. Number of Participants

Up to 42 patients may be enrolled so that approximately 34 patients complete the study. For purposes of this study, a patient completes the study when all scheduled procedures shown in the Schedule of Activities (Section 2) are finished.

If patients discontinue the study before completion of both clamp periods, replacement patients may be enrolled up to 42 patients following agreement between the investigator and the sponsor. The replacement patients will be assigned the treatment sequence of the discontinued patient and complete that sequence in its entirety.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

The use of a crossover design allows each patient to serve as his or her own control, thereby reducing variability. The study is patient- and investigator-blind to minimise potential bias. The Lilly clinical pharmacologist (CP)/Lilly study team will be unblinded.

A euglycaemic clamp technique, the gold standard methodology for assessing insulin action, will be used in this study to provide data on the GD activity of each study insulin lispro formulation. See Section 9.6 for a detailed description of the clamp methodology.

Patients with T2DM were selected based on the need to conduct PK, GD and safety assessments of LY900014 as part of our intended population.

Based on the PK properties of treprostinil (half-life associated with the terminal rate constant in non-compartmental analysis [$t_{1/2}$] = approximately 1 hour) and Humalog ($t_{1/2}$ = approximately 0.7 hours), a minimum of 3 days for the washout period between study dosing for each consecutive study periods and at least 14 days between the last dose of study drug and the follow-up visit are considered appropriate.

5.5. Justification for Dose

Based on previous studies of both insulin lispro (Humalog) and LY900014, the 15 U dose is within the clinical dose range and should provide measurable PK and GD profiles for both study insulins. The safety, PK and pharmacology of LY900014 at similar doses and with similar formulation composition have been assessed in 3 clinical studies in healthy subjects, and in patients with T1DM using MDI or insulin pump treatment and T2DM using MDI. In an MDI regimen, the total daily insulin dose is divided into approximately 50% delivered as the basal insulin dose and the other 50% divided among preprandial insulin doses. In this patient population, the weight-based total daily insulin dose for a large majority of patients is 0.6 to 1.2 U/kg/day, although some highly insulin-resistant patients may require more than 2.0 U/kg/day (Rosenstock et al. 2008). It is estimated that the body weight would be around 100 kg making the total daily prandial dose between 30 and 60 U/day or 10 to 20 U for 3 daily meals. Thus, the 15 U dose of LY900014 within this study would delivery approximately a 0.15 U/kg dose and would not exceed this high estimate for the total daily preprandial insulin.

CCI [redacted]

6. Study Population

Eligibility of patients for study enrolment will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to enrolment. Patients who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

6.1. Inclusion Criteria

Patients are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrolment:

[1] are male or female patients with T2DM for at least 1 year.

[1a] male patients:

No male contraception required except in compliance with specific local government requirements.

[1b] female patients:

A. Women of childbearing potential who are abstinent (if this is complete abstinence, as their preferred and usual lifestyle) or in a same sex relationship (as part of their preferred and usual lifestyle) must agree to either remain abstinent or stay in a same sex relationship without sexual relationships with males.

B. Otherwise, women of childbearing potential participating must agree to use 1 highly effective method (less than 1% failure rate) of contraception, or a combination of 2 effective methods of contraception for the entirety of the study.

a. Women of childbearing potential must test negative for pregnancy prior to initiation of treatment as indicated by a negative serum pregnancy test at the screening visit followed by a negative urine pregnancy test within 24 hours prior to exposure.

b. Either 1 highly effective method of contraception (such as combination oral contraceptives, implanted contraceptives or intrauterine device) or a combination of 2 effective methods of contraception (such as male or female condoms with spermicide, diaphragms with spermicide or cervical sponges) will be used. The patient may choose to use a double-barrier method of

contraception. Barrier protection methods without concomitant use of a spermicide are not a reliable or acceptable method. Thus, each barrier method must include use of a spermicide. It should be noted that the use of male and female condoms as a double-barrier method is not considered acceptable due to the high failure rate when these methods are combined.

C. women not of childbearing potential may participate and include those who are:

   a. infertile due to surgical sterilisation (hysterectomy, bilateral oophorectomy or tubal ligation), congenital anomaly such as mullerian agenesis; or

   b. post-menopausal – defined as either

      1. A woman aged less than 52 years; and being amenorrhoeic for more than 1 year with a serum follicle-stimulating hormone (FSH) level compatible with postmenopausal status; or

      2. Aged at least 52 years and being amenorrhoeic for at least 1 year with a FSH level compatible with post-menopausal status; or

      3. Aged at least 52 years and being amenorrhoeic for more than 1 year

[2] are aged 18 to 70 years at the time of screening.

[3] have a body mass index of 18.5 to 35.0 kg/m^2, inclusive, at screening.

[4] have venous access sufficient to allow for glucose infusion and blood sampling procedures as per protocol.

[5] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures.

[6] are able and willing to give signed informed consent approved by Lilly and the ethical review board (ERB) governing the site.

[7] have clinical laboratory test results within normal reference range for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator.

[8] have a glycated haemoglobin (HbA_1c) <9.0% at screening.

[9] have had no episodes of severe hypoglycaemia in the past 6 months (see Section 9.4.5.3).

[10] are on stable prandial insulin, plus/minus basal insulin (neutral protamine Hagedorn [NPH] insulin, insulin glargine or insulin detemir) with or without a stable dose of metformin, DDP-IV inhibitors, sulphonylureas and SGLT-2 inhibitors, for at least 3 months before screening. See Section 5.1 for OAD washout details.

[11] have a fasting C-peptide ≤1.0 nmol/L. Patients who had C-peptide levels >1.0 nmol/L may be re-tested once. If the re-test yields a C-peptide ≤1.0 nmol/L, then the patient satisfies this criterion.

6.2. Exclusion Criteria

Patients will be excluded from study enrolment if they meet any of the following criteria at screening and/or enrolment:

[12] are investigative site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child or sibling.

[13] are Lilly employees or employees of the investigational site/CRU.

[14] are currently enrolled in a clinical trial involving an investigational product (IP) or any other type of medical research judged not to be scientifically or medically compatible with this study.

[15] have participated within the past 30 days in a clinical trial involving an IP. If the previous IP has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed.

[16] have previously completed or withdrawn from this study.

[17] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study.

[18] have a supine blood pressure at screening outside the range of 90 to 160 mmHg for systolic or 50 to 95 mmHg for diastolic (1 repeat is allowed) as determined by the investigator, or results with unacceptable deviations that are judged by the investigator to be clinically significant for the population, or have a heart rate outside the range of 50 to 90 beats/minute.

[19] have donated blood of more than 450 mL or more in the past 3 months or provided any blood donation within the past 1 month before screening.

[20] have a history or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine (apart from T2DM), haematological or neurological disorders capable of significantly altering the absorption, metabolism or elimination of drugs; of constituting a risk when taking the IP; or of interfering with the interpretation of data.

[21] have known or ongoing psychiatric disorders.

[22] regularly use known drugs of abuse and/or show positive findings on urinary drug screening.

[23] are currently a smoker, used tobacco products on a regular basis in the 6 months before screening, or are intending to smoke during the study period.

[24] have a significant history of alcoholism or drug abuse as judged by the investigator or consuming more than 24 g of alcohol per day for men, or more than 12 g of alcohol per day for women (1 unit of alcohol is defined as 10 mL [8 g] of pure alcohol).

[25] are unwilling to comply with the dietary requirements/restrictions during the study: (i) comply with the fasting requirements of the study, (ii) consume only the meals/snacks provided during the inpatient visits.

[26] show evidence of an acute infection with fever or infectious disease at the time of study entry.

[27] show evidence of human immunodeficiency virus (HIV) infection and/or positive human HIV antibodies.

[28] show evidence of hepatitis C and/or positive hepatitis C antibody (the presence of hepatitis C antibodies in the setting of normal liver function tests and a negative hepatitis C polymerase chain reaction are not an exclusion).

[29] show evidence of hepatitis B and/or positive hepatitis B surface antigen (the presence of antibodies to the hepatitis B surface antigen is not an exclusion).

[30] are women who are pregnant or lactating.

[31] have significant lipohypertrophy in the target abdominal injection area as judged by the investigator.

[32] have a history of renal impairment (exclusion only if estimated glomerular filtration rate [estimated GFR] <60 mL/minute/1.73 m^2 [GFR is estimated according to the Chronic Kidney Disease Epidemiology Collaboration creatinine equation]), or have a serum creatinine level ≥0.000126 mol/L (male) or ≥0.000111 mol/L (female).

[33] have a history of deep vein thrombosis of the leg or repeated episodes of deep leg vein thrombosis in first-degree relatives (parents, siblings or children) as judged by the investigator.

[34] have proliferative retinopathy or maculopathy and/or severe neuropathy; in particular, autonomic neuropathy as judged by the investigator based on a recent (<1.5 years) ophthalmologic examination.

[35] are receiving chronic (lasting longer than 14 consecutive days) systemic or inhaled glucocorticoid therapy (excluding topical, intra-articular and intraocular preparations), or have received such therapy within the 4 weeks before screening.

[36] have, except for current regimen of insulin therapy and concomitant medication(s) (for example, antihypertensive medication, lipid-lowering agent, thyroid hormone replacement medication, hormonal contraception, hormonal replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that treat hyperglycaemia or insulin resistance or that promote weight loss within 14 days before dosing (apart from vitamin/mineral supplements, occasional acetaminophen [paracetamol] or ibuprofen).

[37] have known allergies to treprostinil (CCI [redacted]), insulin lispro, insulin glulisine, related compounds or any components of the formulation, or a history of significant atopy.

[38] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator.

[39] require daily insulin treatment ≥1.2 U/kg/body weight.

[40] are treated with a CSII (insulin pump).

[41] currently receiving insulin degludec therapy or insulin degludec combinations, or have been treated with degludec within the past 30 days.

[42] currently receiving dulaglutide therapy or have been treated with dulaglutide within the past 30 days.

[43] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

Patients will fast for approximately 8 hours prior to dosing until after the glucose clamp procedure is completed, after which patients will receive a meal. Water can be consumed freely during this time. If the clamp procedure is completed in less than 10 hours, then the meal should be delayed until the last PK sample has been taken, unless the investigator deems it necessary to administer the meal for safety reasons. Patients will take their regular prandial insulin with the end-of-study meal and site will record the dose and time in the patients eCRF.

When not a resident at the CRU, patients will be encouraged to follow their normal diets.

6.3.2. Caffeine, Alcohol and Tobacco

No alcohol will be allowed at least 24 hours before each dose and throughout the duration of each CRU visit. Between dosing visits, daily alcohol should not exceed 3 U for males and 1.5 U for females (a unit is defined in Exclusion Criterion [24]).

Patients should refrain from caffeine-containing food/beverages (for example, cola, chocolate, tea and coffee) for at least 12 hours before each dose and throughout the duration of each CRU visit.

Smoking (cigars, cigarettes or pipes) and the use of smokeless tobacco will not be permitted during the study.

6.3.3. Activity

Patients are encouraged to maintain their regular exercise habits; however, they should not undertake vigorous or prolonged exercise within 48 hours prior to each dosing day. After dosing, patients should remain recumbent or sitting in the CRU until the end of the glucose clamp.

Movement will be restricted to retain the integrity of connections to the infusion(s) and the study procedures.

6.3.4. Washout Period

During the insulin-transition and wash-out period hyperglycaemia will be monitored daily by fasting fingerstick blood glucose measured and documented in a diary. Each patient will be instructed to contact the investigational site staff if his/her fasting fingerstick glucose level is >240 mg/dL. The patient will be managed as considered appropriate by the Investigator based on the actual glucose value (e.g., return to the study center for evaluation). If the patient has a fasting fingerstick glucose level >240 mg/dL on two consecutive days, the patient will be instructed to return to the study center the following day to have a fasting plasma glucose performed. If the fasting plasma glucose >240 mg/dL is confirmed, the patient will be withdrawn from the study.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study may be rescreened 1 time. Individuals who still do not meet participation criteria after rescreening (screen failure) may not be rescreened.